

Executive Report: Predictive Analytics for Employee Retention

1. Executive Summary

Employee turnover is a costly challenge for organizations, often leading to increased recruitment costs, reduced productivity, and loss of experienced talent. This project applies machine learning techniques to predict employee attrition and identify key factors influencing employee turnover.

Using a dataset of 1,470 employees, we developed a predictive model to identify employees at risk of leaving and derived actionable insights to support proactive HR decision-making.

2. Methodology & Model Selection

The dataset contains 35 features including employee demographics, job-related attributes, and satisfaction metrics. Since attrition prediction is a binary classification problem with imbalanced data, two models were evaluated:

- Logistic Regression
- Random Forest

Model Comparison:

- Logistic Regression:

Accuracy: ~71.7%

Recall: ~59%

- Random Forest:

Accuracy: ~84.6%

Recall: ~23%

Although Random Forest achieved higher accuracy, it performed poorly in identifying employees who are likely to leave. Logistic Regression demonstrated significantly better recall, making it more suitable for HR applications where identifying at-risk employees is more important than overall accuracy.

3. Key Findings: Drivers of Attrition

Based on Exploratory Data Analysis (EDA) and model insights, the following key factors were identified:

Top 3 Drivers of Attrition

1. OverTime

Employees working overtime show significantly higher attrition rates, indicating possible burnout and workload imbalance.

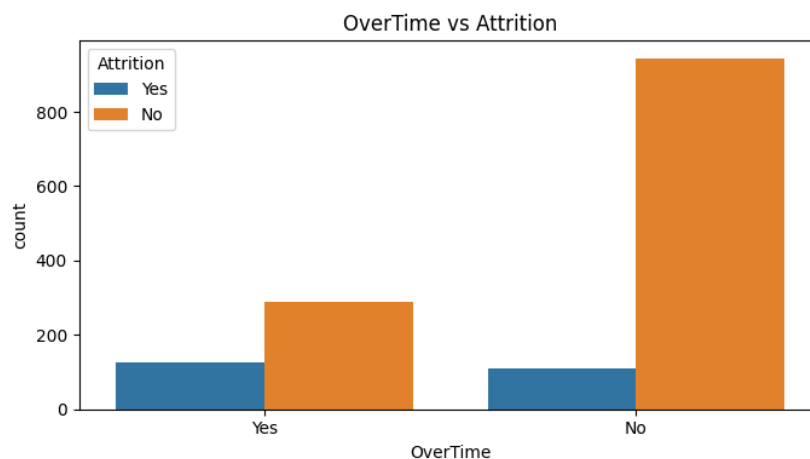


Figure: OverTime vs Attrition

2. MonthlyIncome

Employees with lower income levels are more likely to leave, suggesting that compensation plays an important role in retention.

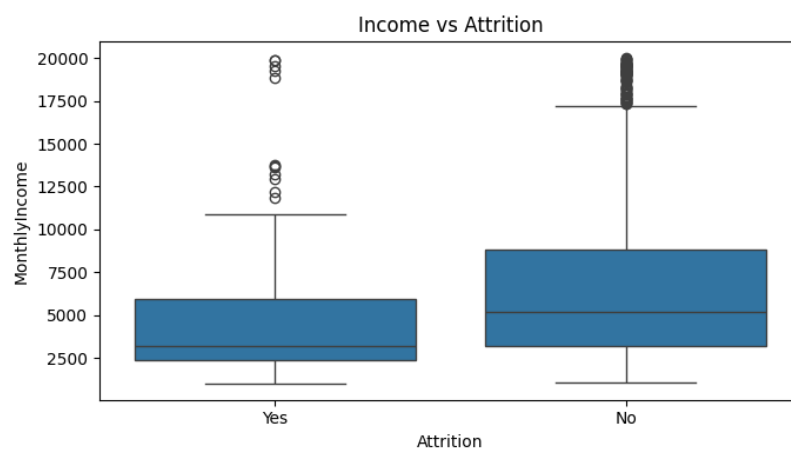


Figure: Income vs Attrition

3. YearsAtCompany / YearsInCurrentRole

Employees who remain in the same role for long periods without growth opportunities are at higher risk of leaving.

Additional Observations

- Employees working overtime have a noticeably higher likelihood of leaving.
 - Income distribution shows that employees who leave tend to have lower salaries compared to those who stay.
 - Certain job roles experience higher turnover than others.
-

Correlation Analysis

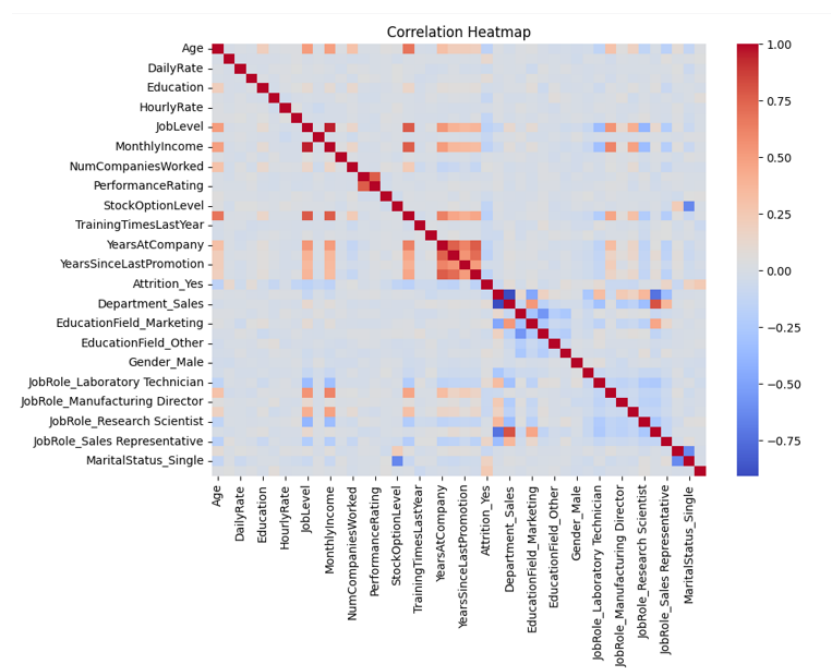


Figure: Correlation Heatmap showing relationships between features

The heatmap illustrates the correlation between different variables in the dataset. Warmer colors indicate positive correlation, while cooler colors indicate negative correlation.

From the analysis, no single feature shows extremely strong correlation with attrition, indicating that employee turnover is influenced by multiple factors rather than a single variable.

However, moderate relationships can be observed between factors such as income, experience, and job-related attributes, suggesting that attrition is a result of a combination of these factors.

4. Department-wise Attrition Analysis

Analysis across departments shows:

- Research & Development and Sales departments experience higher attrition
- Human Resources shows comparatively lower attrition

This suggests that workload, job pressure, and role expectations may vary across departments and influence employee turnover.

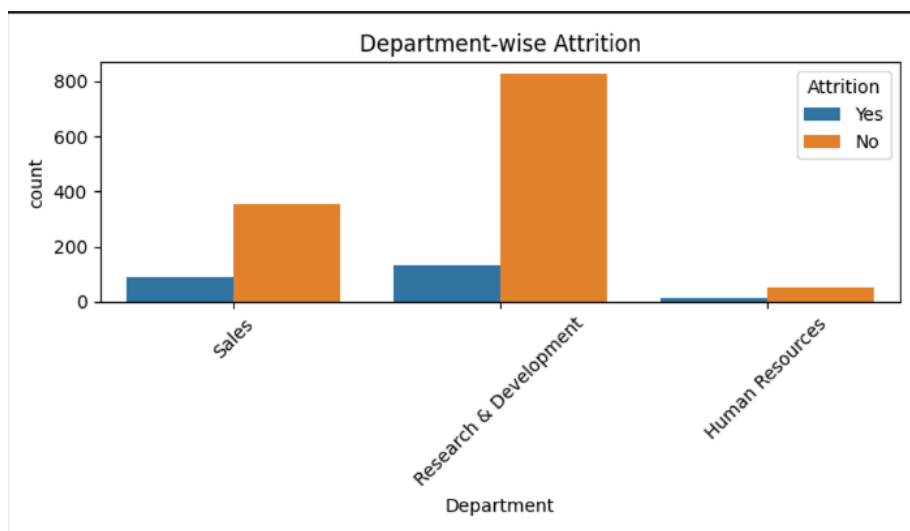


Figure: Department-wise Attrition

5. Risk Profiling

A risk profiling approach was implemented using the Logistic Regression model.

Employees with a predicted probability greater than 0.6 were classified as high-risk employees.

This allows HR teams to:

- Identify potential attrition cases early
- Take preventive actions before employees leave

6. Strategic Recommendations

Based on the analysis, the following actions are recommended:

1. Reduce Excessive Overtime

Monitor workload and redistribute tasks to prevent employee burnout.

2. Improve Compensation Strategy

Review salary structures, especially for lower income groups, to remain competitive.

3. Enhance Career Growth Opportunities

Provide promotions, role changes, and skill development programs to reduce stagnation.

4. Department-Specific Interventions

Focus retention strategies on high-risk departments such as Sales and R&D.

7. Conclusion

This project successfully demonstrates the use of machine learning for predicting employee attrition and identifying its key drivers.

Logistic Regression was selected as the final model due to its superior ability to identify employees at risk of leaving.

The insights derived from this analysis can help organizations move from reactive to proactive HR strategies, improving employee retention and overall organizational